

ELECTROLYTE RELATIONSHIP TABLE

CORRECTED - QC COPY

How the major electrolytes interact and what those relationships mean clinically

↑ increases ↓ decreases ↔ closely related • Opposite = often inverse • Relationships are tendencies; the cause and the patient determine the meaning

RELATIONSHIP	WHAT HAPPENS?	WHY / PHYSIOLOGY	WHAT YOU MAY SEE	NURSING / EXAM CONNECTION
Ca ↔ phosphate OPPOSITE - often	↓ Ca often ↔ ↑ phosphate ↑ Ca often ↔ ↓ phosphate	Phosphate binds Ca. PTH raises serum Ca and increases renal phosphate excretion.	High phosphate can lower Ca → tingling, spasms, hyperreflexia , tetany.	Trousseau : BP cuff → hand spasm. Chvostek : face tap → twitch. Severe: SEIZURE / LARYNGOSPASM risk. Assess neuro + ECG .
Mg ↔ Ca NEUROMUSCULAR / REFLEXES	↓ Mg may cause ↓ Ca → excitability ↑ ↑ Mg → CNS / muscle depression	Both affect nerve/muscle stability. Severe Mg deficiency can impair PTH release or action, lowering Ca.	Low: tremors, cramps, hyperreflexia, tetany. High: ↓ DTRs, lethargy, weakness, low BP/HR, RESPIRATORY DEPRESSION .	Check DTRs , breathing + ECG . LOW = too excitable. HIGH = too relaxed/depressed.
K ↔ acid-base MAINLY METABOLIC	Metabolic acidosis → K tends OUT → serum K ↑ Metabolic alkalosis → K tends IN → serum K ↓	pH and HCO ₃ changes influence K shifts. Respiratory effects are usually smaller.	Either K extreme → weakness + DYSRHYTHMIA risk. Alkalosis may ↓ ionized Ca → cramps/tetany.	WATCH K + ECG . This is a tendency. Cause matters: GI loss can lower K; in DKA, insulin lack is a major reason serum K shifts out.
K ↔ insulin / glucose EXAM 1 - DIABETES	INSULIN → GLUCOSE + K INTO CELLS → serum K can fall	DKA causes osmotic diuresis → total-body K loss. Insulin then moves K into cells.	Serum K may start normal/high, then fall during treatment → weakness + ECG/rhythm changes.	Trend K before/during insulin; monitor rhythm as indicated. SERUM K ≠ TOTAL-BODY K .
Cl ↔ bicarbonate ACID-BASE	Vomiting / NG : lose H ⁺ + Cl → hypochloremic alkalosis; K often ↓. Diarrhea : lose HCO ₃ → metabolic acidosis; Cl may ↑ and K often ↓.	Gastric loss removes acid + Cl. Intestinal loss removes HCO ₃ . Interpret Cl and HCO ₃ together.	Vomiting/NG: weakness, cramps, alkalosis. Diarrhea: dehydration, acidosis; deep/fast breathing may occur.	READ TOGETHER: Cl + HCO₃ + K + FLUID STATUS .
Na ↔ water WATER ↔ BRAIN	Water excess vs Na → dilution → Na ↓ Water deficit vs Na → concentration → Na ↑	Serum Na reflects water relative to Na, not total-body Na. Water moves toward higher osmolality.	Brain cells swell or shrink. Significant changes → confusion, LOC change, SEIZURES .	SODIUM = NEURO + FLUID . Check neuro, I/O, weight. USE CONTROLLED CORRECTION → rapid shifts can cause neurologic injury.
Na ↔ hormones ADH / ALDOSTERONE	↑ ADH : retain water → dilute Na ↓ ↓ / ineffective ADH (DI) : water loss → Na may ↑ Aldosterone : retain Na; water follows; excrete K	ADH controls water retention. Aldosterone promotes Na retention and K excretion.	SIADH → dilutional hyponatremia. DI → high urine output, water loss, hypernatremia risk. Excess aldosterone → K may fall.	Compare Na + K + I/O + urine output + BP + daily weight .
Phosphate ↔ energy ATP / MUSCLE / RESPIRATORY	↓ phosphate → ↓ cellular energy / ATP → muscle function ↓	Phosphate is needed for cellular energy production and muscle function.	General weakness, weak respiratory muscles, poor activity tolerance, ↓ cardiac performance, fall risk.	LOW PHOSPHATE = THINK WEAK . Check breathing strength, mobility, fall/cardiac risk + RENAL FUNCTION .
Mg ↔ K CORRECTION CLUE	↓ Mg → kidney K wasting → low K may persist	Mg deficiency makes it harder for the kidneys to retain K.	K stays low despite replacement; combined deficits increase rhythm risk.	K not improving? CHECK Mg . Monitor ECG + kidney function .
Kidneys ↔ all PUT IT TOGETHER	Impaired regulation / excretion → several abnormalities at once	Kidneys regulate electrolytes, acid-base and fluid. With kidney failure, K, Mg and phosphate may rise; Ca may fall.	Rhythm, reflex, weakness, fluid and pH changes may occur together.	THINK: KIDNEYS + GI LOSSES + MEDS + FLUIDS . Review BUN/Cr, urine output, I/O, meds, orders + trends.

PATTERNS TO RECOGNIZE ON AN EXAM

METABOLIC ACIDOSIS

K tends OUT; effect depends on cause
DKA: total-body K is depleted

METABOLIC ALKALOSIS

K tends IN → K ↓
ionized Ca ↓ → tetany risk

Ca / PHOSPHATE

Often opposite
High phosphate can produce low-Ca findings

LOW Mg + LOW Ca

Excitability ↑
Cramps / hyperreflexia / tetany

VOMITING / NG

H⁺ + Cl lost → alkalosis
K often ↓

DIARRHEA

HCO₃ lost → acidosis
Cl may ↑; K often ↓ from GI loss

INSULIN / DKA

Insulin moves K into cells
Serum K may fall; trend closely

RENAL DYSFUNCTION

K / Mg / phosphate may build up
Check creatinine + urine output

Course basis: NUR 122 electrolyte and acid-base lectures/objectives. Clinical clarifications preserve course intent while distinguishing metabolic K shifts and total-body K loss in DKA.

Review: Lippincott Ch. 10, 17 and 48 • ATI Ch. 45-46 • Exam 1: Days 1-2 exemplars.